

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features Using Tableau

Data Preprocessing Steps

1. Import the Dataset
2. Handle Missing Values
3. Remove Duplicate Records
4. Correct Data Types
5. Handle Outliers
6. Standardize Data
7. Create New Features
8. Filter Irrelevant Columns
9. Validate Data Quality
10. Export Clean Dataset

Business Questions and Tableau Visualizations

1. Distribution of sale prices – Histogram
2. Highest average sale prices by neighborhood – Horizontal Bar Chart
3. Sale prices over time – Line Chart
4. Living area vs sale price – Scatter Plot
5. Highest-priced house style – Bar Chart
6. Bedrooms vs sale price – Box Plot
7. Average sale price by quality – Bar Chart
8. Feature impact on sale price – Heat Map
9. Garage area vs sale price – Scatter Plot
10. Monthly housing sales – Line/Area Chart
11. House age vs sale price – Scatter Plot
12. Average sale price by building type – Bar Chart
13. Sales count by neighborhood – Filled Map/Bar Chart
14. Lot size vs sale price – Scatter Plot
15. Highest average prices by year – Line Chart

Recommended Tableau Dashboards

Dashboard 1: KPIs, Sale Price Trend, Histogram

Dashboard 2: Neighborhood Map, Average Price, Sales Count

Dashboard 3: Living Area vs Price, Bedrooms Box Plot, Heat Map, House Style Chart

Dashboard 4: Monthly Trends, Yearly Prices, Seasonal Analysis

Important Features

SalePrice, OverallQual, GrLivArea, GarageArea, TotalBsmtSF, YearBuilt, YearRemodAdd, Neighborhood, HouseStyle, BldgType, LotArea, BedroomAbvGr, FullBath, HalfBath, GarageCars, MoSold, YrSold

Expected Insights

Premium neighborhoods have higher prices; higher quality and larger homes sell for more; newer homes command higher values; seasonal sales trends exist; garages and basements positively affect prices; price per square foot varies by neighborhood.